

CS1CA Computer Systems Architecture

Memory hierarchy Design (4)

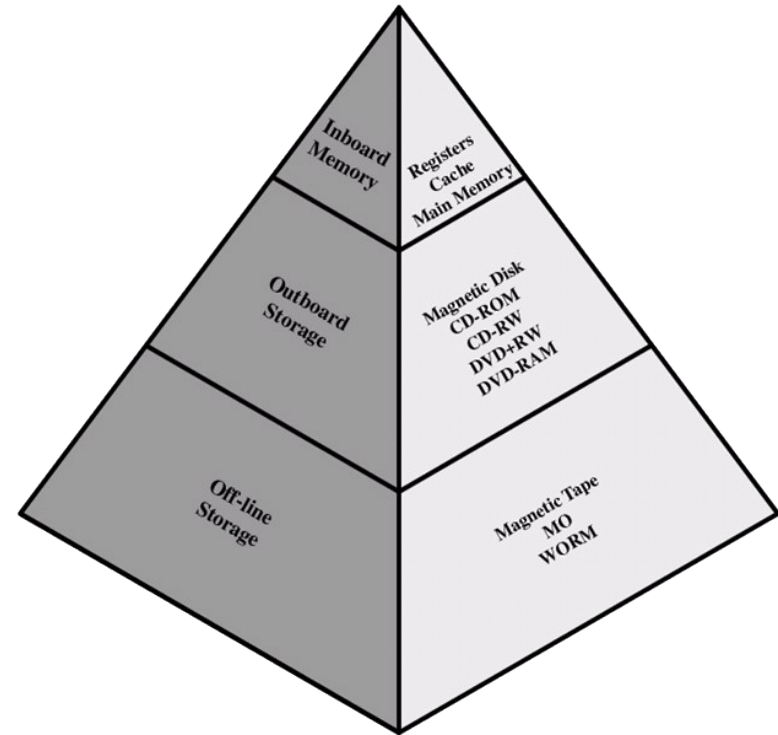
Week 11 lecture 2 of autumn semester 2025

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Continue our discussions on the topic
of memory hierarchy design.

In this lecture

- The following contents will be delivered
 - Main memory: organisations and performance
 - Virtual memory
- The learning outcomes are listed below
 - To gain knowledge of different organisations of main memory
 - To be able to calculate the main memory performance by using *Memory Latency*.
 - To gain knowledge of virtual memory and the four issues of data transfer between main memory and virtual memory.

Main memory: technologies

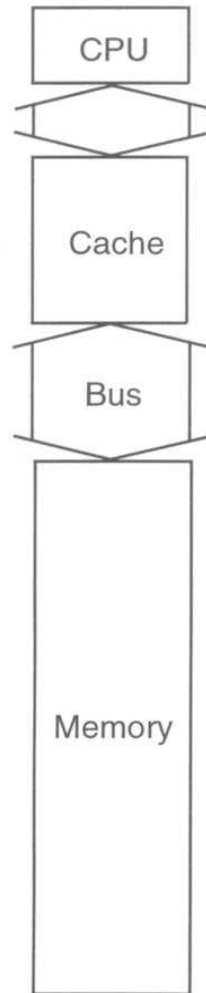
- Memory technology refers to the properties of the underlying hardware mechanisms used to construct the memory system.
- The two main technologies
 - **DRAM**: dynamic random access memory, low cost, used for main memory
 - **SRAM**: static random access memory, higher performance, used for cache
- Embedded processor memory
 - ROM: read-only memory – modification is not allowed
 - Flash memory: nonvolatile and contents alterable
- Other improved technologies.....

Main Memory: performance

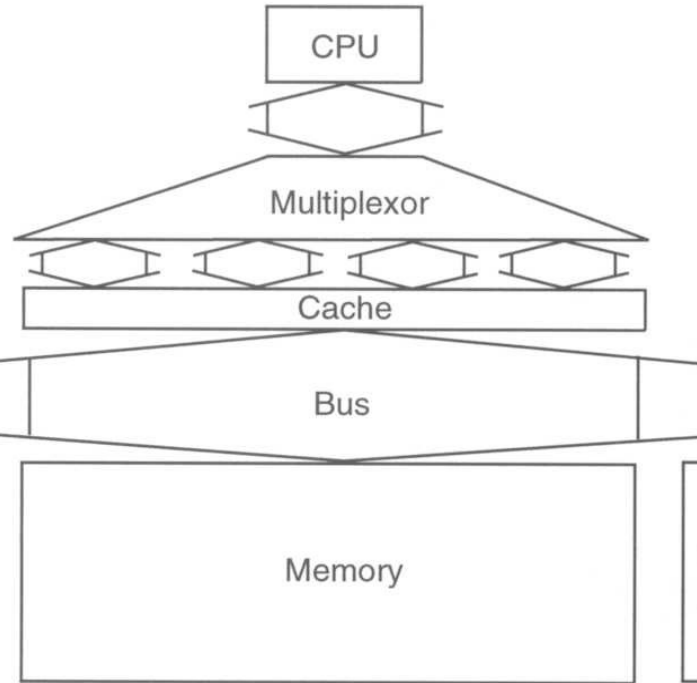
- The performance of the main memory is measured by memory latency associated with memory bandwidth.
- Memory latency can be represented as
$$\text{Memory latency} = \text{Time sending memory address} \\ + \text{Time accessing data} \\ + \text{Time transferring data}$$
- It is easier to improve memory bandwidth with new organisations so that to reduce latency.
- Memory needs address lines to be stable between accesses to ensure the correct address is accessed.

Three organisations for memory systems

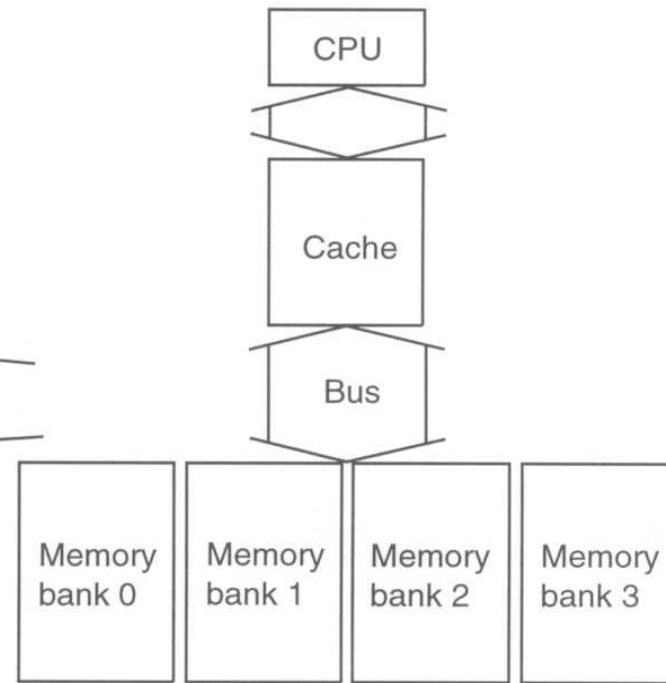
(a) One-word-wide memory organization



(b) Wide memory organization



(c) Interleaved memory organization



Techniques for higher bandwidth - 1

- Wide main memory
 - The main memory is designed with a wide memory bus.
 - The memory bus can be doubled or quadrupled of that used in the basic organisation.
 - ✓ It improves bandwidth hence reducing miss penalty.
 - ✦ The cost for wider connection between the CPU and memory is that a multiplexor has to be added between the cache and CPU.

Techniques for higher bandwidth - 2

- Memory interleaving
 - To take advantage of the potential parallelism of having many DRAMs in a memory system.
 - Divide memory into banks with one-word width so that addresses can be sent to different banks simultaneously.
 - The mapping of addresses to banks affects the behaviour of the memory system.

Techniques for higher bandwidth – 2 (continue)

- Four-way interleaved memory

address	Bank 0	address	Bank1	address	Bank2	address	Bank3
0		1		2		3	
4		5		6		7	
8		9		10		11	
12		13		14		15	

Bank 0 has all words whose addresses MOD 4 is 0

Bank 1 has all words whose addresses MOD 4 is 1

Bank 2 has all words whose addresses MOD 4 is 2

Bank 3 has all words whose addresses MOD 4 is 3

Main memory performance (memory latency) for the basic memory organisation

- **Example:** the given information to the **basic memory organisation** is shown below.
 - 4 clock cycles to send the address
 - 24 clock cycles for the access time per word
 - 4 clock cycles to send a word of data
- The miss penalty (latency of the main memory) for **a four-word (16-byte) cache block** is $4 \times (4 + 24 + 4) = 128$ clock cycles.
- The memory bandwidth:
 $16 \text{ bytes} / 128 \text{ clock cycles} = 1/8 = 0.125 \text{ byte per clock cycle}.$
- Calculate “memory latency” with the given data, under the organisations of wide memory and interleaved memory.

Calculation of memory latency with the given data for the other two organisations

- **Wide memory: for the four word block**

- 4 clock cycles to send the address
- 24 clock cycles for the access time per word
- 4 clock cycles to send a word of data

Memory latency = $4+24+4 = 32$ ccs (clock cycles)

Bandwidth = $16/32 = 0.5$ bytes per clock cycle

- **Interleaved memory: for the four word block**

- 4 clock cycles to send the address
- 24 clock cycles for the access time per word
- 4 clock cycles to send a word of data

Memory latency = $4+24+4*4 = 44$ ccs (clock cycles)

Bandwidth = $16/44 = 0.36$ bytes per clock cycle

Virtual Memory

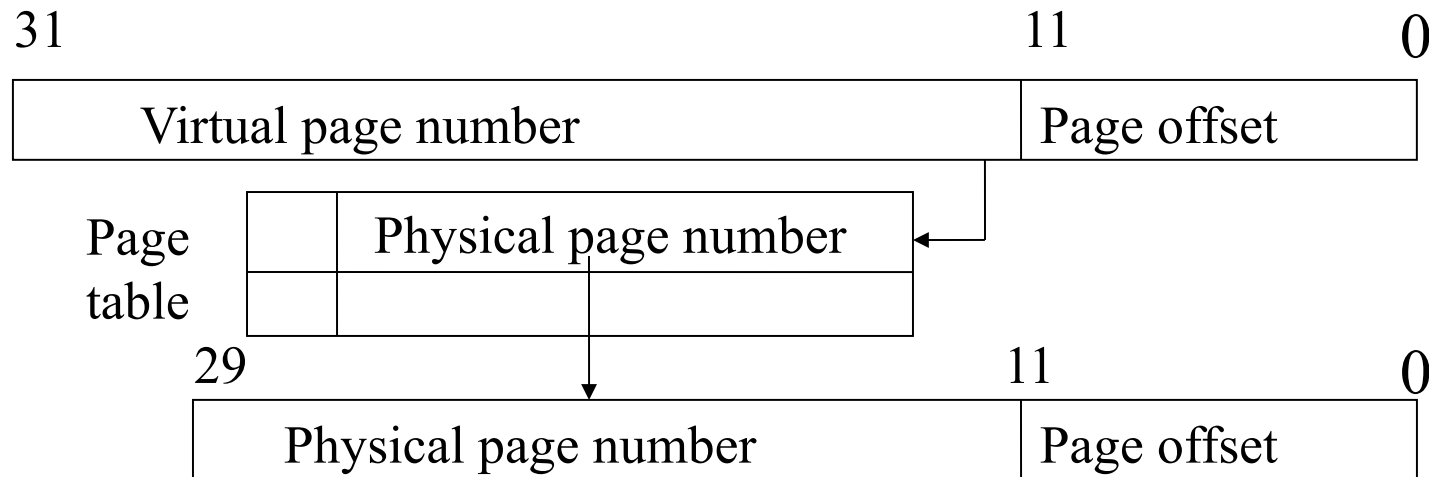
- Computers running multiple processes
 - Each process is with its own address space;
 - It would be too expensive to dedicate the full memory address space to all processes;
 - Many processes use only a small part of their address space – only load this part into the physical memory to start with.
 - Chop programs into pieces (pages or segments)
- Virtual memory reduces time to start a program. Only a small portion of the program is loaded into the physical memory to start it.

Virtual memory: scheme and terms

- A protection scheme restricts a process accessing blocks only belonging to that process.
- Virtual memory divides physical memory into pages or segments and allocates them to different processes.
- Two terms for virtual memory block – a program is chopped into blocks, and represented by
 - Page (fixed size), popularly used; or
 - Segment (variable size)
- Page miss (page fault or address fault): requested page is not found in the main memory.

Virtual memory: address mapping process

- CPU produces virtual addresses that are translated by a combination of hardware and software to the physical addresses which reside in main memory.
- Virtual addresses split into virtual page number (VPN) and page offset (12 bits here);
- VPN is indexed to a page table, where physical page addresses are stored.



Difference between cache and virtual memory

- Replacement on cache misses is primarily controlled by hardware. Virtual memory replacement is primarily controlled by OS;
- The size of processor address determines the size of virtual memory, cache size is independent of processor address size;
- In addition to acting as the lower-level backing store for main memory, secondary storage is also used for the file system (a part of OS) that is not normally a part of memory hierarchy.

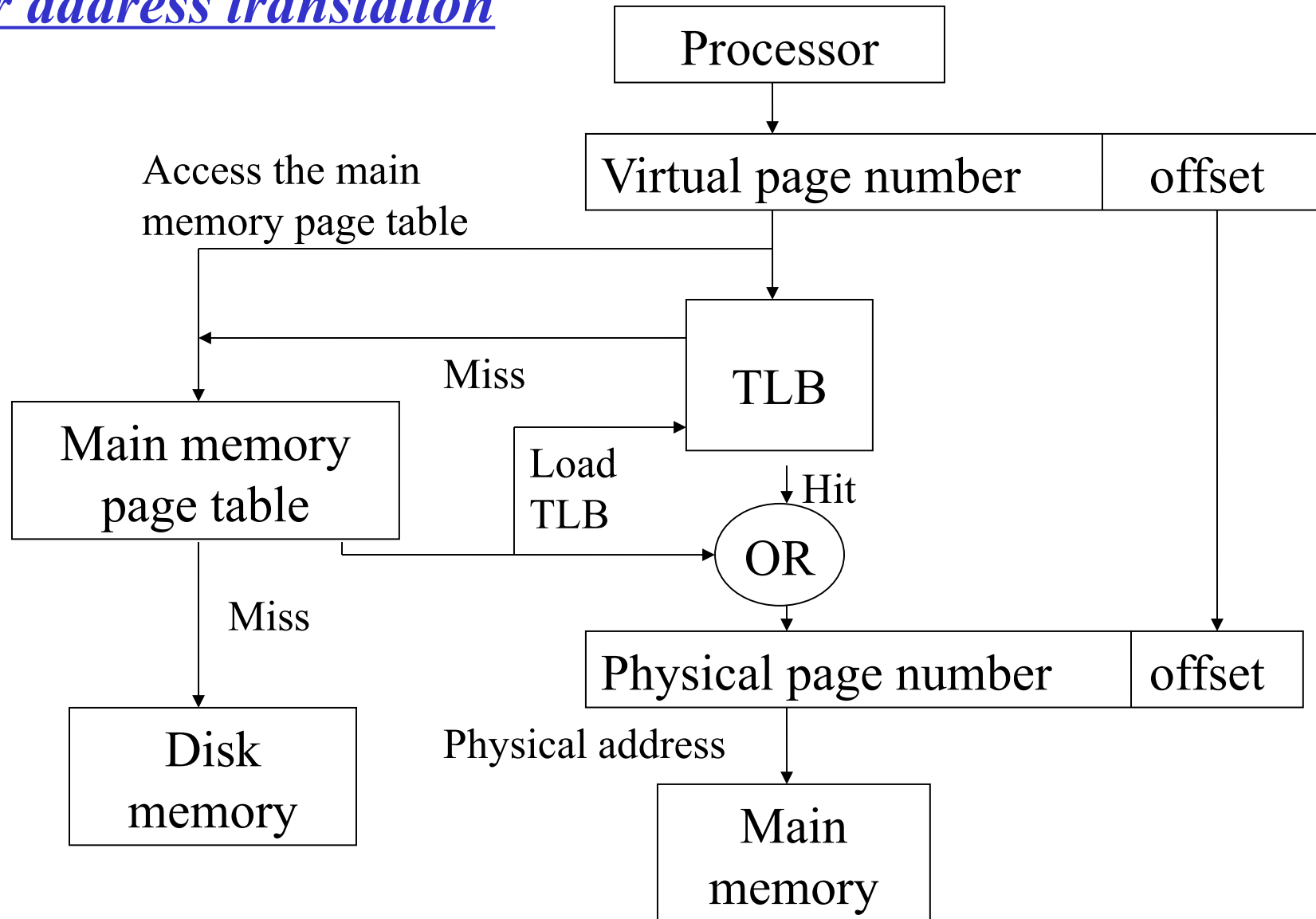
Four issues in virtual memory

- Page placement
 - Any where in the main memory: fully associative
- Page identification
 - A page table indexed by the virtual page number;
 - The size of the table is the number of pages in the virtual address space.
- Page replacement
 - The least-recently used (LRU) page
- Page write
 - Write-back strategy

Virtual memory: fast address translation

- **Problem:** For a large page table in main memory, the table may also be paged. Therefore, every memory access logically takes at least twice as long (first for physical address, second for data).
- **Solution:**
 - Principle of locality: As accesses have locality, address translations for the accesses must also have locality;
 - Keep those address translations, which are most likely to be needed, in a special cache, which is called the **Translation Look-aside Buffer (TLB)**.

TLB for address translation



Virtual memory: page size

- The size of page table (determined by no. of virtual pages) is inversely proportional to page size;
- Larger pages mean a smaller table. Less memory is used for storing the page table;
- Data transfer is efficient for large pages;
- Page faults are less with large page size;
- Small page size means less internal fragmentation, which is the term for unused memory.

Virtual memory: protection scheme

- For each process, set up **a pair of registers** that hold base and bound address. Address is checked to ensure that
 $\text{Base} \leq \text{address} \leq \text{bound}$
- User cannot modify these registers. The operating system must be able to modify these registers so that it can switch process.
- **2 modes for processes**: user mode (application process) and kernel mode (operating system process).
- Add **permission flags** to each page in the page table, e.g. read-only protection.

- Self-learning

Put it all together: the Alpha 21264 memory hierarchy

For the above section:

Please refer to Hennessy & Patterson's book (3rd edition), pp482-489 to review contents of Memory Hierarchy.

End of this lecture

Questions and exercises

- What is the role that the main memory plays in memory hierarchy?
- What is the measure of main memory performance?
- Why do we need virtual memory? What is page or segment in virtual memory?
- Compare and contrast the four issues of data transfer between cache and main memory, and between main memory and virtual memory.
- Explain the workflow of Translation Look-aside Buffer (TLB) in virtual address translation. How does TLB speed up the virtual address translation process?